Technical Summary Sheet 6.5930 / Hardware Architectures for Deep Learning, Spring 2026

Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

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Repository

`final-project/workspace/final_project/` (extends the AccelForge cluster example). Reproduce all numbers with `make study && make notebook && make test` from `final-project/`. The regression suite is the canonical correctness check; twelve figures live under `results/presentation/`.

Three Key Insights / Innovations

1. **The intra-chip mapping is stable; the distributed schedule is what changes.** For the 64-chip TPU cube, AccelForge converges to the same partitioning/dataflow/tile signature for all modeled cube interconnect variants (**0/24** inner mappings change). The benefit lives one layer up: **5/16** workload–scenario pairs flip their optimal communication pattern, and the optimal compiler-visible schedule changes whenever multicast is available. Separating the two layers matches the 6.5930 mapping formalism rather than conflating it.
2. **Hardware multicast and distance-insensitive transport, not raw bandwidth, dominate shared-state workloads.** A 3.5 Tb/s, 6 pJ/bit advanced chiplet baseline—chosen to defeat the bandwidth hypothesis—is still beaten by M1000 by **99.85%** on shared-DRAM multicast: N unicasts collapse to one broadcast, and contention disappears on a switched fabric. The same chiplet is competitive on node-to-node exchange.
3. **Photonics is not a uniform multiplier.** On a 9216-chip Ironwood-class superpod, long-context inference reaches **42.43**× speedup on the Gemma 4 MoE proxy and **14.50**× on the larger long-context Gemini stress proxy. Training improves less: **3.37**× for Gemma 4 MoE and **1.43**× for the larger stress proxy. Strong scaling exposes this compute/HBM limit, while weak scaling shows why flat multicast is valuable for serving more concurrent long-context work.

Headline Results (reproducible from `results/`)

Axis	Value	Setting
16-node energy reduction	71.95%	EVK100 multicast
16-node latency reduction	97.57%	EVK100 multicast
TPU-cube network energy red.	99.93%	EVK100 multicast
TPU-cube AccelForge energy red.	77.45%	EVK100 KV attention
Inner mapping signatures	0/24 changed	TPU cube
Distributed maps changed	5/16 cases	cluster study
Compiler-schedule best gain	73.86%	EVK100 remote streaming
Multicast scaling exponent	0.00	M1000 vs torus / chiplet
Chiplet-relative reduction	99.85%	M1000 multicast
Superpod inference speedup	42.43×	Gemma 4 MoE proxy
Superpod training speedup	3.37×	Gemma 4 MoE proxy
Gemini stress inference	14.50×	Deep Think proxy
Distance model	flat optical	conversion-dominated M1000
Switching sensitivity	OCS/OPS	bulk multicast vs fine-grained traffic

Contributions

Simone Surdi – AccelForge/network model: topology classes (ring, mesh, 3-D torus, optical crossbar / switched fabric); the `LogicalClusterMemory` extension; the iterative network-aware feedback loop (`solve_iteratively`); mapping-signature extraction; cluster, TPU-cube, and chiplet-baseline studies; technology-catalog constants and sensitivity sweeps.
Pranav Sitaraman – Compiler/superpod analysis: schedule enumeration (SPMD collectives, remote HBM streaming, replicated HBM cache, multicast cache fill, MoE locality, full photonic stack); distance-parametrized link model; strong/weak scaling; OCS/OPS sensitivity; the 9216-chip Ironwood-class stage model (prefill / decode / refill / MoE / training / gradient / optimizer); Gemma 4 and Gemini-class proxy workloads; ablation ladder; integration tests; report, poster, and presentation deliverables.

Reproducibility and Threats to Bias

Lightmatter constants are from public product pages and SC25/ISES talks; TPU constants are from Cloud TPU docs and the Ironwood blog / Hot Chips tutorial. EVK100 is reported both as the conservative 3.2 pJ/bit product target and the 1.9 pJ/bit Passage-family sensitivity point; M1000 uses 114.6 Tb/s bidirectional and 2.3 pJ/bit including laser power. The link model separates conversion from distance-dependent transmission: electrical/chiplet links become more expensive with average path length, while photonic links are nearly flat after conversion. Optical multicast is modeled as one logical broadcast (optimistic but explicit). OPS is a sensitivity point, not a Lightmatter product claim. The superpod stage model is analytical with an overlap factor; cycle-accurate simulation is left to future work. Gemma 4 sizes are public; Gemini parameter counts are unpublished, so we use public proxy stress workloads. The regression suite asserts technology constants, distance scaling, multicast behavior, and schedule changes under photonics.